

CS471 – Block 1 Worksheet

Lessons 4-8: Search, Heuristics, and Adversarial Decisions

Block 1 Equations & Concepts Reference

- **Evaluation Function ($f(n)$):** *The total estimated cost of the cheapest path through node n . $g(n)$ is the exact cost to reach n from the start, and $h(n)$ is the heuristic estimated cost from n to the goal.*

$$f(n) = g(n) + h(n)$$

- **Admissibility Condition:** *A heuristic is admissible if it never overestimates the true cost (h^*) to reach the goal. Guarantees optimality in A^* Tree Search.*

$$0 \leq h(n) \leq h^*(n)$$

- **Consistency Condition:** *The estimated cost from n to the goal is no greater than the step cost to a successor n' plus the estimated cost from n' to the goal. Guarantees optimality in A^* Graph Search.*

$$h(n) \leq c(n, a, n') + h(n')$$

- **Dominance:** *A dominant heuristic is more accurate and guarantees A^* will expand fewer or equal nodes. If $h_2(n) \geq h_1(n)$ for all n (and both are admissible), then h_2 strictly dominates h_1 .*
- **Minimax Algorithm:** *Models a zero-sum adversarial game where the MAX agent tries to maximize the utility score while the MIN opponent tries to minimize it.*

$$\max_a \min_{a'} \dots$$

- **Expectimax Algorithm:** *Models stochastic environments where chance nodes dictate outcomes. Instead of assuming worst-case scenarios, the agent maximizes the expected (average weighted) utility.*

$$\max_a \sum_{s'} P(s') V(s')$$

Part I: Guided Practice

Lesson 4. Problem Formulation

An autonomous Combat Search and Rescue (CSAR) helicopter must navigate a 10x10 coordinate grid to reach a downed pilot. Some grid squares contain hostile anti-air threats and cannot be traversed.

- (a) Formulate this search problem by explicitly defining the **State Space**, **Initial State**, **Actions**, **Transition Model**, **Goal Test**, and **Path Cost**.

- (b) What is the difference between a "World State" and a "Search State" in this scenario?

Lesson 5. Uninformed Search

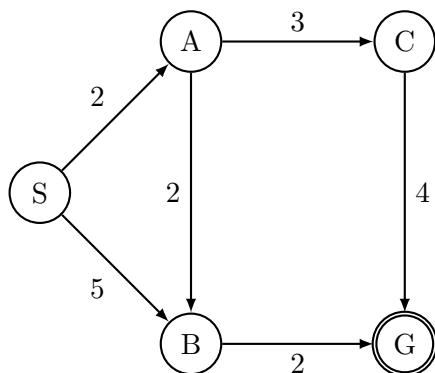
A cyber-agent maps a compromised network. From the entry router (S), it discovers nodes A and B. A connects to C and D. B connects to E. The goal is to locate the adversary's backdoor at node E.

- (a) If the agent uses **Breadth-First Search (BFS)**, in what exact order will the nodes be expanded? What data structure manages the BFS frontier?

- (b) If the agent uses **Depth-First Search (DFS)** (expanding alphabetical ties from left to right), in what exact order will the nodes be expanded? What data structure manages the DFS frontier?

Lesson 6 & 7: Informed Search and Heuristics

An autonomous ground vehicle (AGV) navigates from the Start Base (S) to the Forward Operating Base (G). The map is defined by the following graph. The edge labels represent exact step costs ($g(n)$). The node values in the table represent the heuristic estimate ($h(n)$) to the goal.

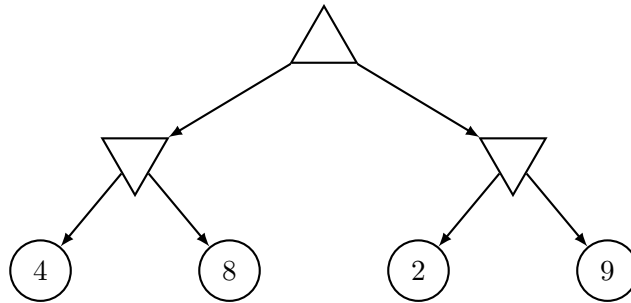


Node	$h(n)$
S	6
A	4
B	1
C	3
G	0

- Execute **Greedy Best-First Search**. List the sequence of nodes expanded. What is the total cost of the resulting path? (Recall: Greedy relies solely on $f(n) = h(n)$).
- Execute **A* Search**. Calculate $f(n) = g(n) + h(n)$ for the frontier at each step. List the final path and its total cost.
- Is the provided heuristic $h(n)$ admissible? Prove your answer by comparing $h(n)$ to the true lowest cost $h^*(n)$ for nodes A and B.

Lesson 8: Adversarial Decisions

A Minimax algorithm models a cyber engagement. You (MAX) are choosing between two defensive scripts. The adversary (MIN) will respond by selecting an attack payload. The leaf nodes represent the final security posture score (higher is better for you).



- (a) Perform the Minimax algorithm. Write the backed-up values inside the MIN nodes and the root MAX node. Which action (Left or Right) should MAX choose?
- (b) If the algorithm evaluates nodes from left to right, using **Alpha-Beta Pruning**, which leaf node(s) will be pruned (not evaluated)? Explain the mathematical logic that triggers the prune.

Part II: Independent Student Practice

Practice 1: Problem Formulation

An AI agent manages power routing for an orbital satellite. The satellite has 3 solar panels that can independently be ON or OFF. The agent's goal is to turn all 3 panels ON. The agent can only toggle one panel per action.

- (a) Define the size of the state space. (How many total possible states exist?)

- (b) Define the exact Goal Test.

Practice 2: Search Algorithms

An intelligence routing algorithm navigates a priority tree. The starting node is 1. Its children are 2 and 3. Node 2's children are 4 and 5. Node 3's child is 6.

- (a) If the algorithm manages its frontier using a **LIFO Stack** (Depth-First Search), what is the exact expansion order to reach node 5?

Practice 3 & 4: A* Search and Heuristics

A loyal wingman drone must navigate a grid. Moving vertically or horizontally costs 1 unit. Moving diagonally costs 1.4 units.

- (a) The drone is at coordinate $(0,0)$ and the target is at $(2,2)$. The agent uses the **Manhattan Distance** heuristic. Calculate $h(0,0)$.

- (b) Is the Manhattan Distance heuristic admissible in this specific environment (where diagonal movement is allowed)? Mathematically prove your answer by comparing $h(0,0)$ to the true cost $h^*(0,0)$.

Practice 5: Stochastic Decisions (Expectimax)

An Electronic Warfare (EW) AI is deciding whether to Jam a radar or Stay Silent. If it Jams, there is a 60% chance the adversary radar reboots (Utility = 10) and a 40% chance the adversary escalates power (Utility = -5). If it Stays Silent, the utility is definitively 2.

- (a) Calculate the expected value (EV) of the "Jam" action.

- (b) According to the Expectimax algorithm, which action should the EW AI take?